BACKGROUND & MOTIVATION

Plant disease and insect pests destroy roughly 40% of global crop production each year and create more than $220 billion in losses, with the burden falling hardest on smallholder farmers. Most existing mobile tools still depend on cloud connectivity that rural fields often lack. AgriVerse addresses that gap with an Android application that runs two convolutional neural networks entirely on-device through TensorFlow Lite: a 38-class plant-disease classifier reaching 95.51% held-out accuracy and a 15-class insect classifier reaching 83.33% test accuracy. Both models use a MobileNetV2 backbone with transfer learning, and their predictions feed a context-aware advisory stack that includes a chatbot, offline encyclopedia content, and satellite, weather, and soil overlays.

THE PROBLEM

of global crop production lost annually from pests and disease pressure

in worldwide annual losses from pests, crop disease, and delayed diagnosis

smallholder farms worldwide (FAO), with 84% operating on under 2 hectares

Smallholder farmers in rural and under-served regions often lack fast, affordable access to agronomists and laboratory diagnostics. By the time disease is identified visually, crop damage may already be severe and treatments may be applied too late or too broadly. Existing diagnosis apps also assume stable internet, which is unrealistic in many fields. AgriVerse is motivated by the need for a practical offline workflow that can diagnose, explain, and guide action without requiring cloud inference.

LITERATURE REVIEW

Prior Work · Plant Disease Classification

CNN-based classifiers dominate agricultural image recognition, but benchmark leaders typically assume cloud inference or heavier deployment footprints.

AlexNet	85.0%	2012	Krizhevsky et al.
GoogLeNet	95.0%	2015	Mao et al.
VGG-16	88.0%	2017	Prajapati et al.
ResNet-50	94.0%	2017	Park et al.
Mohanty CNN	99.35%	2016	Mohanty et al.
MobileNetV2	95.51%	2026	AgriVerse (ours)

All prior entries require cloud inference. AgriVerse runs fully on-device.

Existing Agricultural Apps

Plantix	Large disease database, but cloud-only and subscription-constrained.
Agrio	Rule-based and ML-assisted, though diagnosis still relies on internet access.
Xarvio Scouting	Strong enterprise workflow but oriented toward larger commercial farms.
CropDiagnosis	Image recognition support with limited offline capability and narrower context tools.

RESEARCH GAP

No existing tool combines on-device CNN inference, pest and disease coverage, and contextual agronomic guidance in a single offline-capable package.

HYPOTHESIS

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H1. A lightweight MobileNetV2 trained with transfer learning on PlantVillage can exceed 90% test accuracy when deployed as a TensorFlow Lite model on commodity Android hardware without cloud inference.

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H2. The same backbone can be adapted to a 15-class insect task, and transfer learning plus augmentation can compensate for a comparatively small synthetic training set.

ENGINEERING GOALS

01 **Novel AI Models**

Train plant disease and pest classifiers with measurable accuracy and size targets suitable for real field use.

02 **Offline Deployment**

Convert both models to TensorFlow Lite so inference runs fully on-device with no cloud dependency.

03 **Full-Stack Mobile App**

Ship a unified Android experience that includes camera capture, classification, and advisory views.

04 **Advisory Integration**

Connect model outputs to a GPT-based chatbot, offline encyclopedia content, and environmental overlays.

RESEARCH QUESTIONS

- Q1. Can a MobileNetV2 backbone with frozen ImageNet weights approach parity with cloud CNNs on PlantVillage while staying small enough for on-device deployment?
- Q2. Can transfer learning compensate for a small synthetic insect training set and still produce actionable pest accuracy?
- Q3. What model-size versus accuracy tradeoff remains acceptable for a mid-range Android phone used in rural conditions?
- Q4. Does linking classifier output to an LLM advisor improve diagnosis usability and decision support for farmers?